

What should reconnection remember?

Plastic well changes must persist; elastic remeshing changes must not

Result. An edge flip changes both the local element and its square reference. The plastic part of this change identifies which lattice well the element has reached and should be stored. The accompanying change in elastic deformation belongs to the old parent-child geometry and must remain recoverable. If the full gradient is made exactly invariant, this elastic mismatch is frozen into $H_{\mathcal{F}}$. A later square element can then be reported as elastically strained. The plastic-history construction avoids this: every square geometry is mapped to a square-lattice ground state, while the accumulated plastic well is still remembered.

1 One edge flip is enough

The example uses the pre-flip deformation gradient

$$F^- = \begin{pmatrix} 1 & 1 \\ 0.6 & 1.6 \end{pmatrix}. \quad (1)$$

It can occur in a four-node edge flip for which the descendant associated with this parent is a square triangle after reconnection. Thus the example takes

$$F^+ = I. \quad (2)$$

Figure 1 shows one compatible geometry. The important point is not the particular node positions, but that the parent and child have different elastic parts even though the child is a local energy minimum.

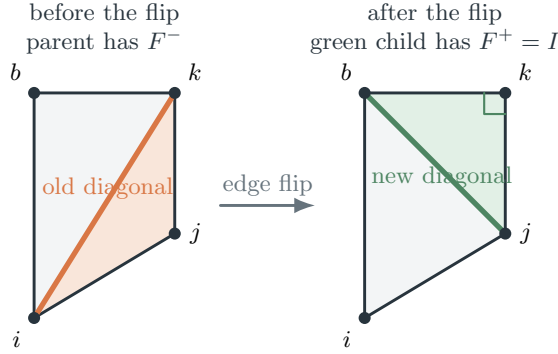


Figure 1: A single asymmetric edge flip. The old highlighted parent is distorted; the associated new child is an exact square reference triangle. The new child should therefore be recognized as a local energy minimum.

2 What is elastic, and what is plastic?

Square-lattice reduction gives the exact multiplicative decomposition

$$F^- = \underbrace{\begin{pmatrix} 1 & 0 \\ -0.4 & 1 \end{pmatrix}}_{F_E^-} \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}}_{F_P^-}. \quad (3)$$

The first factor is a recoverable vertical shear. The second is an integer unimodular lattice transformation:

$$F_P^- = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} =: V_1 H_1, \quad \begin{matrix} V_1: \text{one vertical slip} \\ H_1: \text{one horizontal slip.} \end{matrix} \quad (4)$$

Thus both histories should remember that the element has moved to another lattice well. They differ only in whether the old elastic shear is also made permanent.

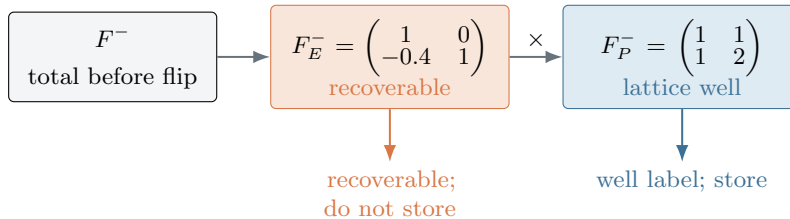


Figure 2: The physical division of responsibility. Elastic deformation is recoverable state; the plastic factor labels the lattice well. Reconnection history should store the blue factor, not the orange one.

3 What each history stores

Let both histories initially equal I . Direct invariance of the full gradient uses

$$H_{\mathcal{F}}^+ = (F^+)^{-1} F^- H_{\mathcal{F}}^-, \quad \mathcal{F} = FH_{\mathcal{F}}. \quad (5)$$

Since $F^+ = I$, this writes the entire old gradient into memory:

$$\boxed{H_{\mathcal{F}}^+ = F_E^- F_P^-}. \quad (6)$$

The plastic-history rule instead uses

$$H_P^+ = (F_P^+)^{-1} F_P^- H_P^-, \quad T_{\text{total}} = FH_P. \quad (7)$$

The square child has $F_P^+ = I$, so only the well change is stored:

$$\boxed{H_P^+ = F_P^-}. \quad (8)$$

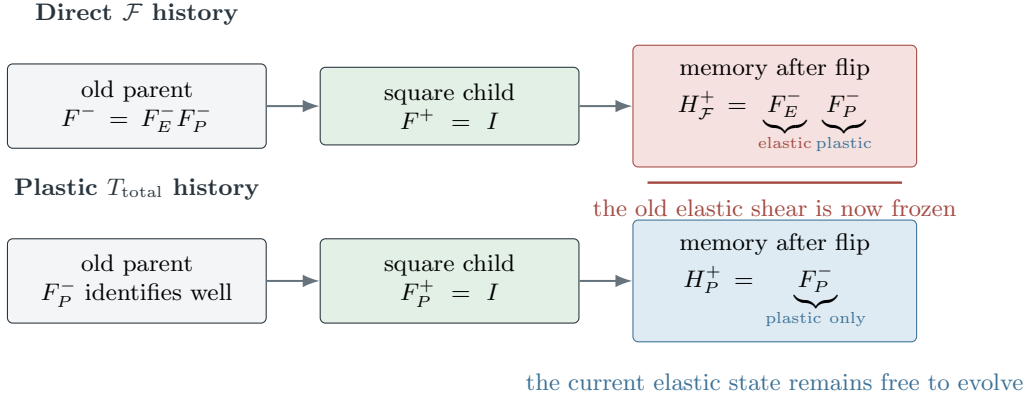


Figure 3: The decisive difference is the content of memory. Both methods remember F_P^- ; only the direct history also stores the elastic parent-child mismatch.

The example $F^+ = I$ makes the mechanism transparent. For a general flip, write

$$F^- = F_E^- F_P^-, \quad F^+ = F_E^+ F_P^+.$$

Then

$$H_{\mathcal{F}}^+ = (F_P^+)^{-1} \underbrace{(F_E^+)^{-1} F_E^-}_{\text{elastic parent-child mismatch}} F_P^- H_{\mathcal{F}}^-, \quad (9)$$

$$H_P^+ = (F_P^+)^{-1} F_P^- H_P^-. \quad (10)$$

Equation (9) shows exactly where the unwanted elastic information enters. If $F_E^+ = F_E^-$, the mismatch is I and the methods agree. If the elastic component changes during remeshing, one flip is sufficient to separate them.

4 Return to a square geometry

Now consider the descendant at any later time when its current local geometry is again a square triangle, so $F = I$. This is a local minimum of the actual element energy. The two reconstructions are

$$\mathcal{F}_{\square} = IH_{\mathcal{F}}^+ = F_E^- F_P^-, \quad (11)$$

$$(T_{\text{total}})_{\square} = IH_P^+ = F_P^-. \quad (12)$$

The plastic result is not literally the matrix I because it correctly remembers one vertical and one horizontal slip. It is nevertheless a square-lattice ground state: after lattice reduction its elastic metric is I . By contrast, reducing $\mathcal{F}_{\square}$ removes F_P^- but leaves

$$F_E^- = \begin{pmatrix} 1 & 0 \\ -0.4 & 1 \end{pmatrix}, \quad (F_E^-)^T F_E^- = \begin{pmatrix} 1.16 & -0.4 \\ -0.4 & 1 \end{pmatrix} \neq I. \quad (13)$$

Thus $\mathcal{F}$ would describe a non-minimal elastic state even though the current nodes form a minimal-energy square triangle.

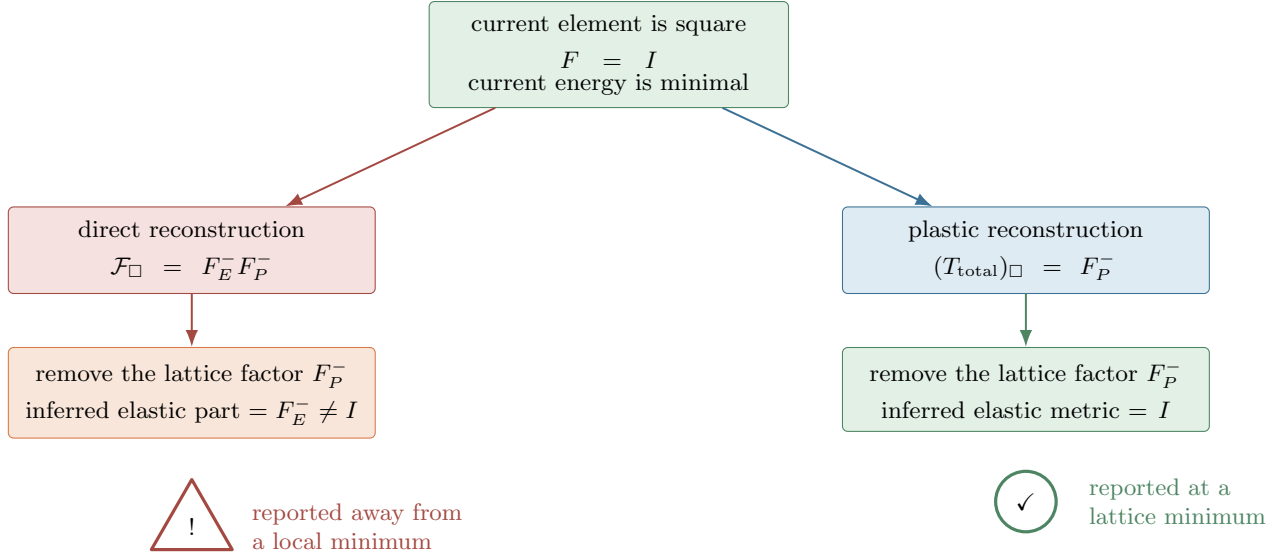


Figure 4: The same square current geometry receives two different interpretations. The plastic history retains the correct well while allowing the elastic state to return to a minimum. Direct invariance leaves the old elastic shear embedded in the reconstructed gradient.

5 Interpretation and recommendation

	Direct $\mathcal{F}$ history	Plastic T_{total} history
What survives a flip	Plastic well plus elastic parent-child mismatch	Plastic well only
What remains free to recover	Only deformation occurring after the mismatch was stored	The complete current elastic component
Square current geometry	Can be reported outside the local energy minimum	Always maps to a square-lattice ground state
What is remembered correctly	Exact algebraic invariance of the inherited full parent gradient	The accumulated sequence of lattice slips
Appropriate interpretation	A record of an imposed parent-child correspondence, not a reliable current elastic state	Reconnection-invariant plastic history combined with the current elastic state

The history variable is an approximation necessitated by reconnecting; it does not define the simulation forces. Its physically useful role is therefore to remember information that the current geometry cannot recover on its own. The current geometry already determines the elastic deformation. What it loses at an edge flip is the discrete lattice-well label.

Consequently, the history should preserve F_P but not the change in F_E . The recommended reconstructed measure is

$$T = F_P H_P, \quad T_{\text{total}} = F_E T = F H_P, \quad H_P^+ = (F_P^+)^{-1} F_P^- H_P^-. \quad (14)$$

This construction remembers every plastic well change while ensuring that a minimal-energy current geometry is always represented by a minimal-energy deformation gradient.